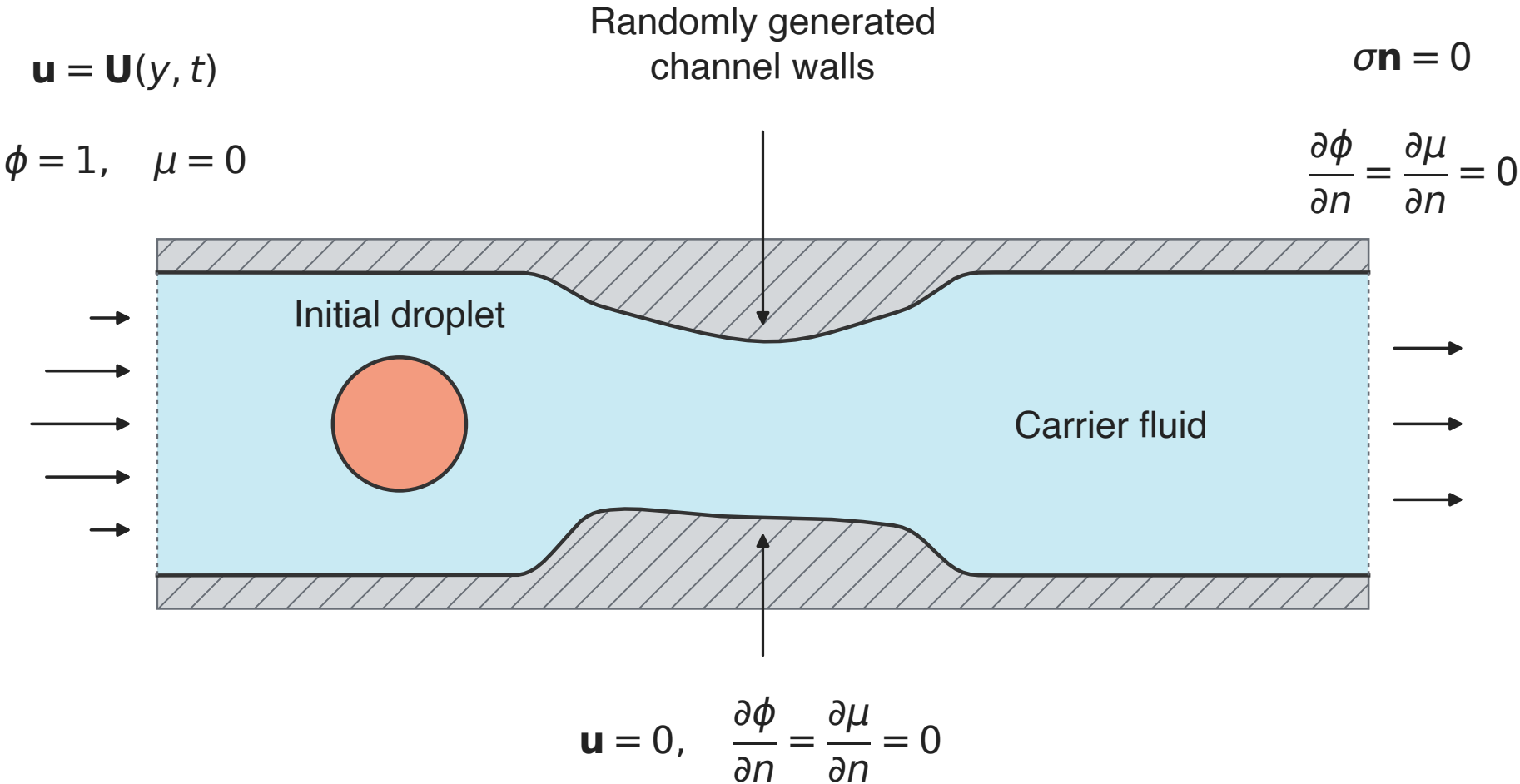

Randomly generated
channel walls
$\mathbf{u} = \mathbf{U}(y, t)$
$\sigma\mathbf{n} = 0$
$\phi = 1, \quad \mu = 0$
$\frac{\partial \phi}{\partial n} = \frac{\partial \mu}{\partial n} = 0$
Initial droplet
Carrier fluid
$\mathbf{u} = 0, \quad \frac{\partial \phi}{\partial n} = \frac{\partial \mu}{\partial n} = 0$